NCERT • GRADE 7 • SCIENCE

Digestion in Animals

C09 • Life Processes in Animals

Mouth • Food pipe • Stomach

Small intestine • Large intestine

Chapter 9 digestion companion notes

Textbook solutions first • Visual understanding • Revision

Original student notes based on NCERT. All illustrations are editable LaTeX drawings.

Part A • NCERT Chapter 9 textbook questions (pp. 135–136)

How these notes fit the collection

The existing **C09_Respiratory_System.tex** explains breathing and respiration. These companion notes teach the chapter's **nutrition and digestion** section and answer all 10 exercise questions here as a complete reference. See the blue boxes for the digestion questions.

Q1 • Journey of food through the alimentary canal

Food ⟶ **Mouth** ⟶ **Food pipe (oesophagus)** ⟶ **Stomach** ⟶ **Small intestine** ⟶ **Large intestine** ⟶ **Rectum** ⟶ **Anus.** Remember that the liver and pancreas help digestion but are *not* stops along the alimentary canal.

Q2 • Iodine with chapati and potato

A (unchewed chapati): blue-black: starch present.
B (chewed chapati): lighter blue-black, or no blue-black after sufficient chewing: saliva breaks down some starch into sugars.
C (boiled mashed potato): blue-black: potato contains starch.
Remember: iodine tests starch, not sugar. The exact result for B depends on chewing time and the amount of starch remaining.

Q3 • Role of the diaphragm

(iii) It helps in inhalation and exhalation. It moves down during inhalation to increase chest volume and moves up during exhalation.

Q4 • Match respiratory parts to functions

(i)–(a), (ii)–(d), (iii)–(e), (iv)–(b), (v)–(c).
Nostrils: air enters; nasal passages: hair/mucus trap dust; windpipe: conducts air; alveoli: gas exchange; rib cage: protects lungs.

Q5 • Is breathing the same as respiration?

Ask Anil: *"Does inhaling air itself release energy from food? Where is glucose broken down?"* Breathing is the physical movement of air; cellular respiration uses oxygen to help break down glucose and release energy.

Q6 • Do we inhale oxygen or air?

Anu and Tanu are scientifically correct, but **Tanu is more precise** in describing the oxygen-rich inhaled air. Shanu's claim is incomplete because air is a mixture of gases, mostly nitrogen, not pure oxygen.

Q7 • Why do we sneeze in dusty air?

Dust irritates the lining of the nose. Sneezing is a protective reflex that helps force out irritants. Tiny nasal hairs and mucus also trap dust.

Q8 • Why did Anusha breathe faster?

Possible reasons include **(1)** more intense running/effort, **(2)** different fitness or recent rest, **(3)** individual breathing differences, or **(4)** illness/heat. Exercise requires more oxygen and more removal of carbon dioxide.

Q9 • Rice flour, saliva and iodine experiment

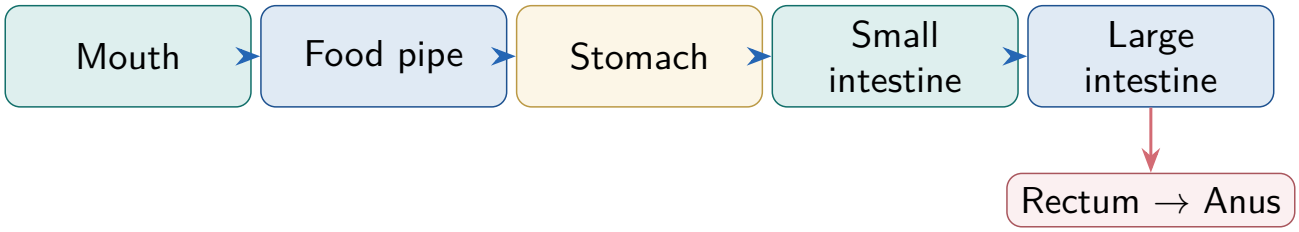
Yadu tests whether **saliva breaks down starch**. Tube A (without saliva) gives blue-black with iodine; tube B (with saliva) gives a weaker colour or none if sufficient starch has been digested. Both tubes must be left under comparable conditions.

Q10 • Lime water and exhaled air

Rakshita investigates whether **exhaled air contains more carbon dioxide** than surrounding air. Bubble equal amounts of air into equal volumes of fresh lime water; the exhaled-air tube turns milky faster because CO_2 causes this reaction. *Safety:* use separate clean equipment; the teacher should supervise.

Part B • Digestion explained visually

A journey through the body:



The four essential processes

Ingestion: food enters. **Digestion:** complex food breaks into simpler substances. **Absorption:** digested nutrients enter blood. **Egestion:** undigested waste is expelled. Egestion is *not* the same as excretion of metabolic wastes.

Part C • What happens at each stage?

1. Mouth • start of digestion

Teeth crush and chew: **mechanical digestion**. Saliva moistens food and helps break starch into simpler sugars: **chemical digestion**. The tongue mixes and pushes softened food towards the oesophagus.

2. Oesophagus and stomach

The food pipe moves food by wave-like **contraction and relaxation**. The stomach churns food and secretes digestive juice, acid and mucus. Digestive juice starts protein digestion; acid helps digestion and kills many germs; mucus protects the stomach lining.

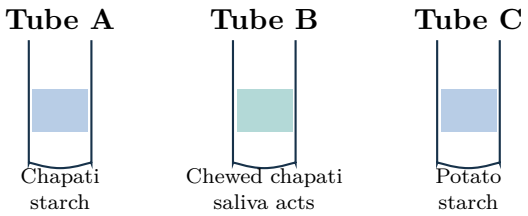
3. Small intestine • main absorption site

Bile from the **liver** helps neutralise acid and breaks fat into small droplets. **Pancreatic juice** helps digest carbohydrates, proteins and fats. The intestine's own juice completes digestion. Tiny finger-like projections in its wall increase area for nutrient **absorption into blood**. The small intestine is roughly six metres long in the textbook.

4. Large intestine, rectum and anus

The large intestine absorbs remaining **water and some salts**. Semi-solid stool is stored in the **rectum** and leaves through the **anus**. Fibre-rich food helps stools pass more easily.

Part D • Read an experiment like a scientist



Before iodine: pale mixtures. **After iodine:** A and C blue-black; B often less blue-black because saliva has acted on starch. Keep water, time and temperature alike for a fair test.

Part E • Animals have different digestive adaptations

Cow vs bird

Cow / buffalo: ruminants partly swallow grass and later bring it back to the mouth for further chewing (*rumination*).
Bird: food is ground in a muscular *gizzard*, sometimes with swallowed grit. Many birds do not have teeth.

Part F • Common mistakes and revision

- **Wrong:** "The liver is where food passes next." **Right:** the liver supplies bile; food passes through the alimentary canal.
- **Wrong:** "The small intestine is shorter than the large intestine." **Right:** *small* refers to width, not length.
- **Wrong:** "Iodine detects saliva." **Right:** iodine detects *starch*.
- **Wrong:** "Only the stomach digests food." **Right:** digestion begins in the mouth and is completed in the small intestine.

60-second self-check

1. Where is most digested nutrient absorption? **Small intestine**.
2. Which tube takes food from mouth to stomach? **Oesophagus**.
3. Which organ produces bile? **Liver**.
4. Which organ stores stool? **Rectum**.
5. Name the process of bringing partially digested grass back for chewing. **Rumination**.